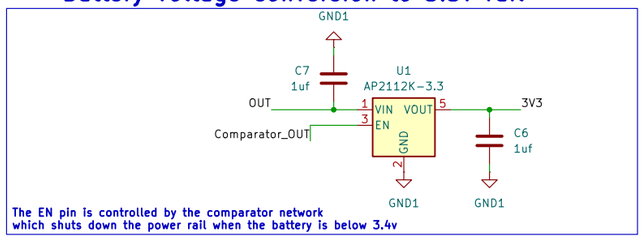
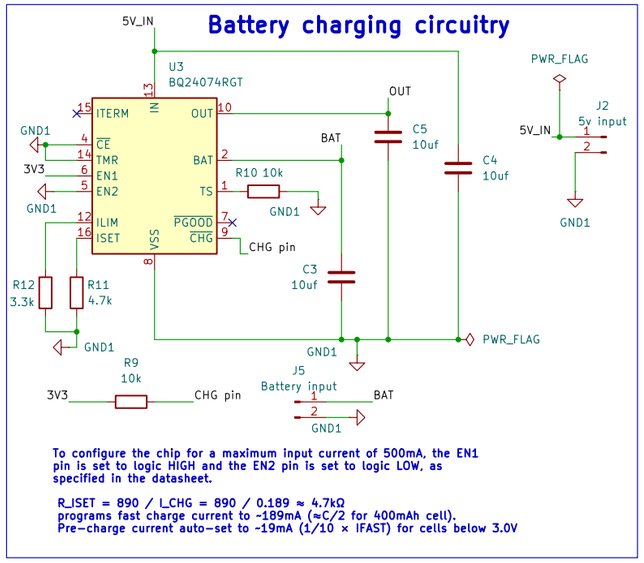
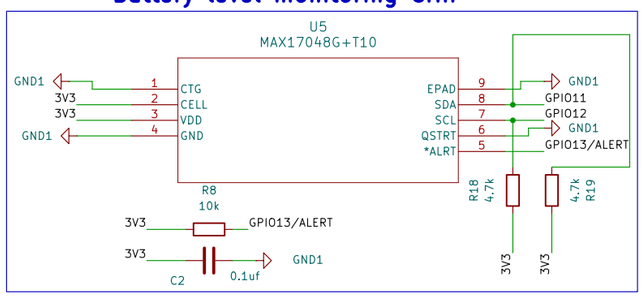
Battery voltage conversion to 3.3v rail



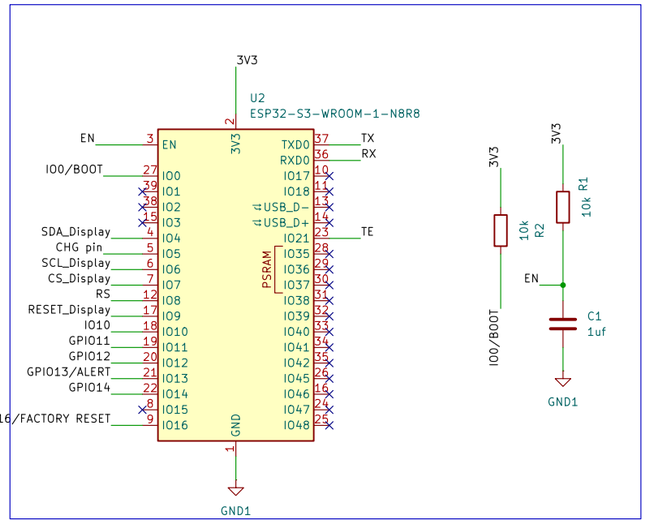
Battery charging circuitry



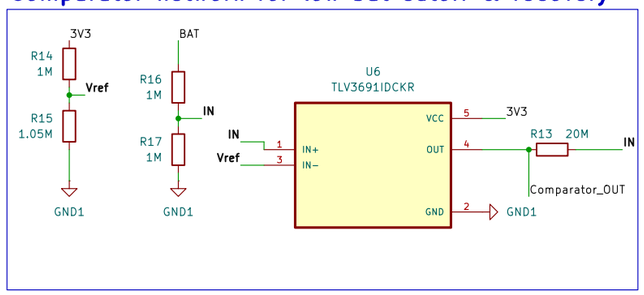
Battery level monitoring CHIP



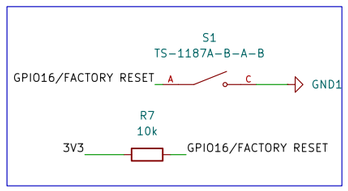
MCU



Comparator network for low bat cutoff & recovery



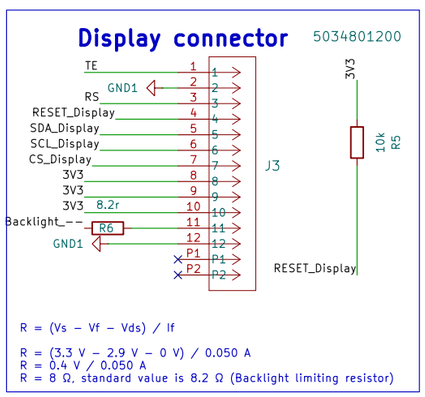
Factory reset button



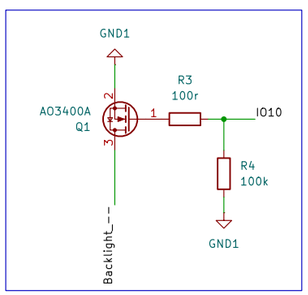
ESP32-S3-WROOM-1-N8R8 GPIO Pin Mapping

Category	MCU GPIO Pin	Net Label	Function
Power	VCC (3.3V)	VCC_3V3	3.3 V power rail
Power	GND	GND	Common ground
Programming / Boot	EN	RESET	MCU enable / reset
Programming / Boot	GPIO0	BOOT	Boot mode selection
Programming / Boot	GPIO43	UART_TX	Programming / Debug TX
Programming / Boot	GPIO44	UART_RX	Programming / Debug RX
Display (ST7789)	GPIO4	SDA_Display (MOSI)	SPI display data
Display (ST7789)	GPIO6	SCL_Display (SCLK)	SPI display clock
Display (ST7789)	GPIO7	CS_Display	Display chip select
Display (ST7789)	GPIO8	RS (D/C)	Display data / command
Display (ST7789)	GPIO9	RESET_Display	Display reset
Status & Control	GPIO5	CHG_Status	BQ24074 charging status input
Status & Control	GPIO10	Backlight	Backlight drive output
Status & Control	GPIO13	MAX17048_ALERT	Fuel gauge interrupt
Status & Control	GPIO16	FACTORY_RESET	Factory reset push button
I^2C Bus	GPIO11	I2C_SDA	MAX17048 SDA
I^2C Bus	GPIO12	I2C_SCL	MAX17048 SCL

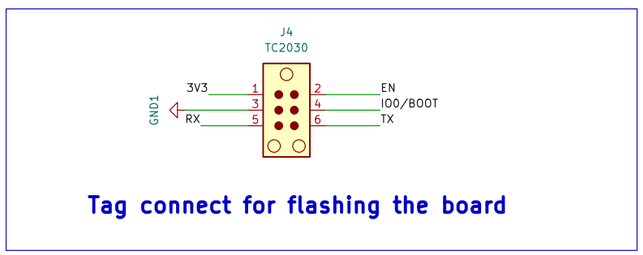
Display connector



Backlight driver circuit



Tag connect for flashing the board



Linear Resonant Actuator (Vibration motor)

